
The Computer Graphics Book Of Knowledge

History of Computer Graphics (CG)

Computer Graphics (CG) was first created as a visualization tool for scientists and engineers in government and corporate research centers such as Bell Labs and Boeing in the 1950s. Later the tools would be developed at Universities in the 60s and 70s at places such as Ohio State University, MIT, University of Utah, Cornell, North Carolina and the New York Institute of Technology.

The early breakthroughs that took place in academic centers continued at research centers such as the famous Xerox PARC in the 1970s. These efforts broke first into broadcast video graphics and then major motion pictures in the late 70s and early 1980s. Computer graphic research continues today around the world, now joined by the research and development departments of entertainment and production companies. Companies such as George Lucas's Industrial Light and Magic are constantly redefining the cutting edge of computer graphic technology in order to present the world with a new synthetic digital reality.

1940s

The very first computer assisted graphics began in many different unrelated fields around the world. There is a very blurred line that is crossed somewhere between mechanical and analog computer assisted graphics, and the first directly digital computer generated graphics that would associate with today as being true CG.

The very first radiosity image.

While at MIT in the 1940s, Professors Parry Moon and Domina Eberle Spencer were using their field of applied mathematics to calculate highly accurate global lighting models which they called *interreflection*. The illumination algorithms were based on those by H. H. Higbie, published in his 1934 book, *Lighting Calculations*.

Lacking any display or output mechanism, the image itself was created by painstakingly selecting Munsell paper samples that matched the output data of their mathematical model. The paper was cut out and ironed together by hand to create the image shown here in print for the first time in over 50 years.

[IMAGE OF THE RADIOSITY PIC]

(The original image is still hanging in the office of Dr. Domina Spencer at the University of Connecticut.)

The images were first presented at the 1946 National Technical Conference of the Illuminating Engineering Society of North America, and published two years later (in color) in the book: *Lighting Design* by Moon, P., and D. E. Spencer. 1948. (Addison-Wesley. Cambridge, MA) The book was used for many years to teach lighting theory at MIT in the architecture curriculum there. Dr. Spencer went on to teach at Tufts, Brown, Rhode Island School of Design, and the University of Connecticut where she remains active today.

1950s

€John Whitney Sr. devises his own computer assisted mechanisms to create some of his graphic artwork and short films. One of his sons John Jr. works with and learns from his father from childhood through high school.see biography

€Pioneering artists Stan VanderBeck, Michael Noll and others at Bell Labs in New Jersey created computer assisted graphics using analog computer devices and plotter output. Later, in the mid 1960s, digital computers and film recorders would be used to produce some of the earliest CG animated films

€Bill Fetter experimented with early vector graphic CAD at Boeing (Seattle) in the late 1950s using an IBM 7094 computer with punch card input and a Gerber plotter.

1950

Artist Ben Laposky uses analog computers to help him create oscilloscope artwork.

1951

Vectorscope-type graphics display on the Whirlwind computer at MIT. A device similar to a light pen allowed direct input to the screen.

The General Motors Research Laboratory begins to study the role of computer aided graphical design applications. (This would later result in the development of the DAC-1)

1955

SAGE system at MITs Lincoln Lab uses the first true light pen as input device. (Bert Sutherland)

1956

Lawrence Livermore National Labs connects graphics display to IBM 704; use film recorder for color images

Bertram Herzog at the University of Michigan Computing Center uses analog computers to generate CRT graphic studies of military vehicle behavior.

1957

1st image-processed photo at National Bureau of Standards. (By whom? Why?)

The IBM 740-780 (paired with a separate IBM 704 computer system) generated a sequence of points on a CRT in order to represent lines or shapes. Time lapse film photography was used to capture the images as they were drawn on the screen.

The Defence Departments Advanced Research Project Agency (ARPA) is founded.

1958

MITs Lincoln Labs: Funded in part by the Air Force; Steven Coons, Ivan Sutherland, and Timothy Johnson begin

working with the TX-2 computer system to manipulate drawn pictures. Ivan Sutherland later began refining the work into his famous Sketchpad system while a student at MIT. DEC later commercialized the TX-2 as the PDP-6.

1959

The first commercial film recorder the General Dynamics Stromberg Carlson 4020. (Produced in San Diego, CA.)

1959

DAC-1 (Design Augmented by Computers): First computer aided drawing system. Created by Don Hart and Ed Jacks at General Motors Research Laboratory and IBM. (Not unveiled until the Fall Joint Computer Conference in Detroit in 1964.) The system was originally based upon a IBM 7090 computer (later upgraded to a 7094 in 1963) augmented with extra disc space and a specially designed IBM 7969 image processing system. Input was with punch cards, but was also capable of scanning in drawings. The final data could be output to either 35mm film (by way of a CRT), a hard copy plotter, or used to drive computer controlled machining devices.

Biography: John Whitney Sr. (1917-1995)

A Los Angeles native, Whitney was a pioneer in many forms of experimental and abstract art before turning to computers to aid in his graphic creations. He attended Pomona College in California in the 1940's and was the first in a wave of artists to begin new techniques of computer assisted graphics.

The integration of analog computer controlled camera and artwork were at first more a pioneering use of motion control than of computer graphics. In point of fact, the devices these early artists used were not even thought of as computers, being more akin to analog music synthesizers. From his experience working in the aircraft industry during World War II, Whitney realized that components of a computerized anti-aircraft controller could be used to drive his mechanisms. One of his sons (John Jr.), recalled buying the state-of-the-art M7 anti-aircraft control computer at surplus. The still unopened crate was 12 feet long, 7 feet wide and seven feet tall. These synchronized mechanisms would ultimately be used to calculate abstract shapes, and change them over time to create beautifully abstract forms and animation.

In the 1950s Whitney worked in Hollywood as an animation director at UPA, most notably contributing graphic elements for the Saul Bass designed opening credits to Hitchcock's Vertigo. Whitney then founded Motion Graphics Inc. in 1960 and produced animation for both television and film, devising the slit scan technique for his early short film Lapis. This technique would later be made famous when used by Con Pederson as a portion of the famous StarGate ending sequence of Stanley Kubrick's 2001: A Space Odyssey.

In 1966 with the help of a grant from IBM and a Fortran programmer named Jack Citron, Whitney made his first digital computer short film called "Permutations". His next works: Matrix 1 and Matrix 2 were done at Cal Tech, followed by Matrix 3 at Triple-I in 1971. It was at this time that he met Larry Cuba who would later be asked by Whitney to collaborate with him in 1975 on his last 16mm film project Arabesque, funded in part by an NEA grant. Both Whitney and Cuba would work briefly at Robert Abel's effects company before digital computer graphics were begun there.

Beginning in the mid 1980's, a new collaborator Jerry

Reed(sp?) translated Cuba's Fortran code into Pascal for use on new personal computer hardware that Whitney could use at home. Whitney continued to create abstract computer animation on his own with the aid of this new PC technology that freed him from the reliance of large company owned mainframe machines and the need for sponsored grants. His work would be displayed in galleries on the same PC hardware he created it on. His last commercially available collection of works, called *Moondrum*, was released on video in the late 1980s. Today his son Michael Whitney is serving as archiver for his father's work, and recently organized a retrospective showing at UCLA.

1960's

Computer assisted graphics were being created more widely as a new and unique art-form by people such as Charles Csurik, Ken Knowlton and John Whitley Sr.

Many pioneering artistic films and artworks were created at Bell Labs from about 1962 to 1967 by artists and programmers such as E.E.Zajac, Leon D. Harmon, Ken Knowlton, A.Michael Noll, Lilian Schwartz, M.R. Schroeder and Stan Vanderbeek. An IBM 7094 computer ran a Stromberg-Carlson 4020 film recorder, programmed in FORTRAN to run Ken Knowlton's Beflix animation system.

Much of the nation-wide university computer science research conducted at the time was due in part to funding from the government's "Advanced Research Project Agency" (ARPA). ARPA at the time took a very hands-off approach to funding. This allowed researchers an un-preserved environment in which to concentrate on the work, without the heavy bureaucracy, paperwork and political constraints more common today. Much to the benefit of researches was Ivan Sutherland who headed ARPA for a time. With good funding, little oversight and many brilliant young minds inspiring each other, it was a unique and special time that produced the very foundation of today's computer graphic tool sets.

Herb Freeman had a school of CG development going on at NYU including Alvy Ray Smith in his first professor's job out of Stanford in 1969. Freeman and his students had already solved the hidden-line problem, a very big deal at the time.

[Quote] Also on the pixel side of things, Azriel Rosenfeld at the Univ of MD, and Ron Baecker was developing some of the very first computer animation. I saw his system GENESYS at an NYU demo in the early 1970s which means Ron probably did the development (in Canada, Toronto, I think) in the late 1960s. -Alvy Ray Smith

Nicholas Negroponte teaches Computer Aided Design (CAD) at M.I.T in the mid to late 60s, and develops the URBAN5 system. A light pen allows interaction directly on the CRT, in combination with keyboard instructions. Points and symbols are added in orthographic mode with a perspective option entered after the fact in order to view structures three-dimensionally. An intelligent system study, URBAN5 was abandoned by 1968 in favor of other projects.

The Society for Information Displays is formed in the early 60s, publishing papers dealing mostly with military applications.

At this same time, practical commercial and industrial use of computer graphics begins to take hold in many areas of

design and manufacturing. Throughout the decade at Boeing, William Fetter and Robert Woodruff (Computing Technology Administrator) leads many important industrial applications of vector generated CG.

€Architectural and urban planning programs (typically written in FORTRAN on machines like the IBM 1130 or 1800) are used at the firm †Skidmore, Owings & Merrill, in Chicago and in the University of Texas School of Architecture. A sample workstation would consist of a Rand tablet providing input, with output to pen plotters such as the Calcomp.

€In the late 60s, the Electronics Laboratory of General Electric (Syracuse, NY) produces a prototype visualization system for NASA and the Office of Naval Research. The system produced real-time color raster graphics on a monitor as a training aid to astronauts going to land on the moon. This same system was used by Prof. Peter Kamnitzer of the UCLA School of Architecture and Urban Planning to simulate urban development plans.

Biography: Dr. Dave C. Evans (1924-1998) MORE

One of a very few who could be called a true "founding father" of computer graphics, Dave Evans is perhaps best known for being the co-founder of "Evans & Sutherland Computer Corporation". Mr. Evans was at one time chairman of the computer science departments at both the University of California Berkeley and University of Utah, where he started the venerable doctoral program that would give birth to so much of the foundation of our industry. Evans first associated with Ivan Sutherland at both Berkeley and the Pentagon's "Advanced Research Project Agency" (ARPA).

Mr. Evans made many contributions to a wide range of computer technologies, and a great many of his students went on to flourish in the brand new field of computer graphics, becoming true pioneers themselves. Students of Mr. Evans include Alan Kay (Co-founder of Xerox PARC), Jim Clark (founder of both Silicon Graphics and Co-founder of Netscape Communications), John Warnock (Co-founder of Adobe Systems) and Edwin Catmull (see biography in Programming chapter).

Dave Evans passed away on Oct.3rd, 1998.

1960

William Fetter of Boeing coins the term "computer graphics" for his human factors cockpit drawings. With help from Walter Bernhardt, and others, Fetter input an aircraft drawings coordinates into a database and plotted out a calculated perspective on a †Illustromat 1100, plotter.

John Whitney Sr. founds Motion Graphics, Inc. in LA.

1961/62

Spacewar: The first popular computer graphic game written by students Steve Russell, Slug Russell, Shag Graetz, and Alan Kotok of MIT to run on the DEC PDP-1. (DEC's PDP-1 cost \$120,000 and MIT's was one of only 50 ever built) The large round CRT display featured graphics controlled by primitive handmade joysticks. The object being to maneuver away from a gravitational †sun, force at the center, and avoid the other enemy ships, while trying to blast him with your own space torpedoes!

The original source code (which ran on 4k of memory!)

can still be found at
www.media.mit.edu/groups/el/projects/spacewar/sources
 or ftp://ftp.digital.com/pub/DEC/sim/sources/sim_2.3d.tar.Z
 There's also a copy of the PDP-1 manual at
www.dbit.com/~greeng3/pdp1/pdp1.html

1962

"Sketchpad: A Man-Machine Graphical Communication System" is presented by Ivan Sutherland as his Ph.D. thesis at the M.I.T. Lincoln Laboratory. The user could input simple lines and curves by drawing directly on the screen with a light pen. The computer, the TX-2, had a whopping 320 kilobytes of memory and a 9 inch monochromatic CRT.

While Sketchpad was strictly 2D, a few years later Timothy Johnson expanded its capabilities into three dimensions as Sketchpad 3. The display CRT was divided up into the now familiar four views, top front side and perspective.

ARPA J.C.R. Licklider is put in charge of the new Information Processing Techniques Office (IPTO) at the Defense Departments Advanced Research Project Agency (APRA). The initial \$14 million dollar budget supported projects at MIT, Berkeley and Carnegie-Mellon.

1963

Biography: Ivan Sutherland

Born in 1938, Hastings, Nebraska; Ivan Sutherland is truly an early founding father of computer graphics. After completing his Ph.D. at M.I.T. (where he developed Sketchpad) in 1963, Ivan Sutherland joined the army and was assigned to the NSA as an electrical engineer. One year later, he was transferred to the Defense Department's Advanced Research Projects Agency (ARPA, later DARPA), and given responsibility for the newly-established Information Processing Techniques office. At age 26, Lt. Sutherland was given a secretary and \$15 million a year, and was told to "go sponsor computer research." Which he gladly did for the following two years until joining the faculty at Harvard late in 1966. It was here with student Bob Sproull that they developed the Head Mounted Display (HUD), for remote viewing; the first Virtual Reality.

In 1968 Ivan formed the Evans & Sutherland company with partner Dave Evans. Ivan was now a part time tenured professor at the University of Utah, where Evans was the founding head of the Computer Science Department. Dr. Sutherland had first met Evans during a visit to U.C. Berkeley as part of his ARPA work.

Ivan's last research in computer graphics was a paper titled: "A Characterization of Ten Hidden-Surface Algorithms," by Sutherland, Sproull and Schumacker. The paper solved many of the largest problems of the day in this critical area of rendering and display technology.

Later, as co-founder (with Carver Mead) and head of the Department of Computer Science at California Institute of Technology from 1976 - 1980, Dr. Sutherland developed and promoted courses involving integrated circuit design, the seed of knowledge that helped create the Silicon Valley industry.

In the early 1980s at Carnegie Mellon University, Ivan did some research on a six legged walking robot, large enough to carry a driver. (And controlled by a joy stick acquired by brother Bert from his contacts in the Navy as a former fighter pilot!)

In 1980, Ivan and Bob Sproull had started the consulting

firm I Sutherland, Sproull & Associates. Sun bought the company in 1990, which then became the nucleus of Sun Microsystems Laboratories.
Today?

Charles Csuri created an analogue computer and used it to make transformations of a drawing. He completed a series of drawings based upon the paintings of old masters such as Durer, Goya, Ingres, Klee, Mondrian and Picasso

Ken Knowlton's programs BEFLIX and EXPLOR are used to create early computer films at Bell Telephone Labs.

The 1st computer art competition, sponsored by Computers and Automation magazine.

The Spring Joint Computer Conference has several people from MIT presenting papers on graphical display technology: Steven Coons, Ivan Sutherland, Tim Johnson, Bob Stotz, Doug Ross and Jorge Rodriquez.

John Lansdown pioneered the use of computers as an aid to architectural planning, making perspective drawings on an Elliott 803 computer in 1963, modelling a building's lifts and services, plotting the annual fall of daylight across its site and authoring his own Computer Aided Design applications.

Edgar Horwood developed a computer graphic mapping system used by the U.S. Housing and Urban Development. HUD publishes I Using Computer Graphics in Community Renewal.

Frieder Nake at The Computer Institute of the Stuttgart Polytechnic uses the Graphomat Zuse Z 64 Drawing machine to produce 4 color plotter drawings.

1964

[QUOTE] I did my first computer graphic at the Physical Sciences Lab at New Mexico State University. I was asked to generate an equiangular spiral antenna for one of the early Nimbus weather satellites. The old engineers asked me, a student, to do the tedious hand-drawing. I got a computer to draw the spiral quickly, amazing the old-timers. Alvy Ray Smith

Ivan Sutherland (a recent MIT gradute) takes over at the Information Processing Techniques Office (IPTO) at ARPA. It is suggested by his predisesor J.C.R. Licklider to take on a 'deputy', Bob Taylor. (The office's budget would reach \$30 million by 1969, when it was changed to DARPA the Defense.) Sutherland transitions out of his office by early 1966 to go to Harvard, leaving Bob Taylor in charge. (Bob Taylor would later go on to play a key role in staffing the famous Xerox PARC.)

1965

Dr. David Evans founds the Computer Science Department at the University of Utah

Ohio State University CG program started by Charles Csuri.

1st computer art exhibition, at Technische Hochschule in Stuttgart

Bella Julesz and A. Michael Noll exhibit for the 1st U.S. computer art exhibition, at Howard Wise Gallery in New York (April, 1965)

196?

First commercially available graphics computer: IBM 2250
(When was the DEC 338??)

[FACTOID COSTS] A typical graphic display CRT cost about \$40,000 US. Rand input tablets are about \$10,000 US, and Calcomp plotters about \$4000 US.

1966

"Odyssey": The first consumer computer graphics games product by Ralph Baer of Sanders Associates. Later marketed at Magnavox.

Permutations: With a grant from IBM and a Fortran programmer named Jack Citron(sp?), John Whitney Sr. made the first digital computer short film. An IBM 2250 Graphic Display Console created dot patterns which were then recorded onto black and white 35mm film. The filmed images were then further enhanced with a specially designed optical printer to add secondary motion and color.

As Associate Professor at Harvard, Ivan Sutherland and his students, Bob Sproull, Jim Clark and others, took earlier "Remote Reality" vision systems of the Bell Helicopter project, and turned it into what we now call Virtual Reality by replacing the camera with computer images. The first such computer environment was no more than a wire-frame room with the cardinal directions -- North, South, East, and West initialed on the walls. The viewer could "enter" the room by way of the West door, and turn to look out windows in the other three directions.

Affectionately called *The Sword of Damocles*, because of its ceiling mounted gear, what they called the "Head-Mounted Display," later became known as Virtual Reality.

The International Conference on Design and Planning: *Computers in Design and Communication*, is held at the University of Waterloo (Ontario). Organized by Professors Constant and Krampen of the Design Department, it was brought together to enlighten and inform designers of emerging computer technologies.

1967

[QUOTE] *At the same time that geometry-based computer graphics (CG) was being invented so was sampling theory-based computer graphics, often called image processing (IP) or imaging. In the early days, two conferences - one for each half of the discipline - would be held side by side. One of the earliest journals was called Journal of Computer Graphics and Image Processing. Its editors were Herb Freeman and Azriel Rosenfeld (CG and IP, respectively). The earliest paper that I actually have in possession on IP side is *Processing of Tiros Cloud Cover Pictures on a Digital Computer*, by Albert Arking, 1967, but I'm sure the literature is much older. It's easy for the geometry based guys to leave all this stuff out and vice versa.* Alvy Ray Smith

Allen Bernholtz and William Warntz of the Laboratory for

Computer Graphics and Spacial Analysis at Harvard University use computer graphics to study layout and sound patterns for hospital floorplans.

Cornell University's School of Architecture is founded by Professor Donald Greenberg.

Charles Csuri creates his famous 'Hummingbird' film. A ten minute long, vector interpolated 16mm film animation that is later purchased by the Museum of Modern Art as part of their permanent collection.

2D morphing techniques used were started by Les Mezei at the University of Toronto

The MIT Center for Advanced Visual Studies is founded by Gyorgy Kepes

The Computer Technique Group in Tokyo Japan is funded at the IBM Scientific Data Center. Engineers and designers create many beautiful and varied computer graphic art works, using image processing and geometric transformations. Members include Koji Fujino, Junichiro Kakizaki, Masao Komura, Fujio Niwa, Makoto Ohtake, Haruki Tsuchiya, and Kunio Yamanaka.

Stephen Coons is Associate Professor of Mechanical Engineering at M.I.T., where he heads the computer aided design (CAD) group. He invents a method for patch continuity

1968

Robert Mallary, Professor at the Department of Art at the University of Massachusetts developed the TRAN2 computer program for calculating three-dimensional sculpture

Cybernetic Serendipity: The Computer and the Arts exhibition at London Institute of Contemporary Arts (ICA) is organized by Jasia Reichardt. The first major public computer art show, Cybernetic Serendipity is also a book published at the same name.

The UK's Computer Arts Society (CAS) is founded by John Lansdown at the Royal College of Art.

The EVENT ONE computer art exhibition is held at the Royal College of Art. Chaired and organized by John Lansdown.

CalComp (California Computer Products) holds a competition for the best 'Computer Plotter Art', with scholarship and cash prizes.

The first computer animation in the UK was the FLEXIPEDE made by Tony Pritchett. Made at the Open University.

Several computer art publications are available in Europe including Bit International out of Zagreb, and Page by the London Art Society, a monthly magazine which actually lasted until the mid 80s.

Ivan Sutherland joins the Computer Science Department at University of Utah

The very first computer graphics company was formed by two of the leading researchers of the day, Drs. David C. Evans and Ivan E. Sutherland. Aptly named Evans & Sutherland, it provided a vector system comprised of custom designed hardware and software previously available only to one of a kind, multi million dollar military sites.

Dicomed is founded as a manufacturer of hardware and software products to apply computer graphic technology to the field of medical radiology. Their systems operate by scanning x-ray films, converting the information into digital data, enhancing it and redisplaying the processed image. (See their web site at www.dicomed.com). Still in business 30 years later, providing professional high resolution digital image capturing technologies.

Bill Fetter contributed to the first (vector based) computer generated television commercial in 1968 while at Boeing.

1969

[IMAGE RAM 2/9 plotted drawing 1969]

Edward Zajec begins a long career of fine art aided by the computer, creating plotter output works using an IBM 60/20 at Carlton Collage in Minnisota. He would later spend 10 years as an Artis-In-Residence at the University of Triesse in Italy. He returned to the united states to Syracuse University in 1980 to reinvigorate the CG program there which had begun in the early 70s. <http://web.syr.edu/~ezajec/ez-plain.html>

[IMAGE RAM 3/16 plotted drawing 1969]

[COINCIDENCE!] It should be noted that this Edward Zajec (with an *le*.) is not the same as the Edward Zajac (with an *la*.) who worked at Bell Labs. Two early pioneering CG artists, two very closely spelled names!

LDS-1 (Line Drawing System). The first commercial CAD wireframe graphics machine system released by E&S. Incorporated hardware design from Garry Watkins, designed input by Chuck Seitz (University of Utah faculty 1970-73), Bob Shumaker and others.

[LDS-1 FACTOID] A local play-on-words for the LDS-1 was based on the fact that the Mormon church was very prominent in Utah, and more commonly known by the contraction of the Church's full name *Latter Day Saints* .: LDS

John Warnok (University of Utah Ph.D. 1969) Developed the Warnock recursive subdivision algorithm for hidden surface elimination.

Alan Kay (University of Utah Ph.D. 1969) First developed the notion of a graphical user interface with the Alto project at Xerox PARC (Palo Alto, CA), which directly influenced the

design of Apple MacIntosh computers.

Computer artist Lloyd Sumner creates Christmas cards under the company name Computer Creations.

Bell Labs developed the first frame buffer for storing and displaying 3bit images.

Gary Demos first becomes acquainted with computer assisted graphics with John Whitney Sr. who is teaching at Cal Tech in California. An IBM 2250 ran a custom operating system, images were photographed in Ektachrome and printed on Kodachrome.

1970's

Widespread commercial use of this early technology did not begin until the 1970s when early pioneers saw the potential in the broadcast video market for the new creative tools. Companies like Image West(LA), Dolphin Productions (New York) and Computer Image Corp (Denver, President Lee Harrison) used these realtime computer assisted video graphics machines to introduce new imagery to both broadcast clients and the viewers at home.

[BIO SNIP] Lee Harrison, the inventor of analog video-based computer animation, was the founder of Computer Image Corporation(1969) in Denver, CO.; where the ANIMAC, Scanimate, C.A.E.S.A.R., and System IV analog animation devices were developed. Lee won an Emmy for SCANIMATE in 1972.

Relatively affordable commercial random access frame buffers became available in the mid to late 70s which opened up the market for CG production. The input for these earliest machines were often banks of patch wires, paper tape or punch cards, very different from today's mouse and graphic interfaces.

These first million dollar commercial machines were mostly capable of only limited, video resolution raster based graphics. While their output was limited in most cases to videotaping or filming monitor screens, their imagery did introduce the public at large to the new art form. By the end of the decade affordable raster technology out paced the earlier vector graphic mainstay.

€Pioneering work done by Jim Blinn at the Jet Propulsion Laboratory (JPL) in Pasadena California (started in 1975 by Bob Holzman). David Em (who would work with Alvy Ray Smith at Xerox PARC on Dick Shoup's Superpaint system in about 1974 or so) also later joined Jim at JPL to create some of the early serious computer art in raster form.

€Nelson Max at Lawrence Livermore National Laboratories uses CG to illustrate basic biologic research; the first scientific visualizations.

€Jim Kajiya, Gary Demos, Steve Gabriel and the Cal Tech contingent

[FACTOID] Artist and Author Jasia Reichardt estimates in 1970 that there are perhaps 1000 people in the world working with computer graphics, who are not involved in pure research or mechanical design. (In other words: CG artists)

1971

Gary Demos visits NASA AMES and Evans & Sutherland while researching a documentary film about computers for 1Dimension Films, in LA. It is there that he first meets Ivan Sutherland and expresses his ambitious desire to create complex and realistic high resolution CG images for films. (Gary is only about 21 years old at the time) Since most of the hardware and software technology that would make this possible does not yet exist, Gary joins E&S in hopes of creating these missing pieces. John Warnock ran the San Jose E&S office before going to NYIT, and Ivan himself was working on his own hidden surface solutions at the time. Gary helped develop a high precision 1data table, (table not tablet because it was 4 feet by 5 feet) accurate to 100th of an inch for digitizing images. The table used two pens to define two simultaneous points in 3D space. Programming was done in assembly code on a PDP-11 with a Picture System 1 for vector display.

Both Henry Gouroug and Bui Toi Phong worked on shading at E&S, so that area was well covered needless to say. Gary and the E&S team next tackled the challenge of building the first ever random access frame buffer. They began with the first 8 DRAM chips ever produced, which came from a company in Texas called Mostek(sp?).

1972

PONG developed by Nolan Bushnell. (Later founder of Atari)

The first feature film appearance of CG: West World. A "block pix" scene done at Information International Inc. (III; aka "Triple I") Led by John Whitney Jr., digitally processed film was used to portray a pixelated android point of view.

1973

ACM/SIGGRAPH is formed

'Interact' at the Edinburgh Festival, a seminal event in establishing the use of computers for the creation of art works. Organized by John Lansdown.

Edwin Catmull (Ph.D. 1974 University of Utah) develops both the Z-buffer algorithm and the concept of texture mapping in 1973-74. (Texture mapping techniques were later refined by Catmull, Alvy Ray Smith, Tom Duff, Lance Williams, and Paul Heckbert at NYIT.

First physical structure designed entirely with computer-aided geometric modeling software: A large Easter egg which is still standing in Vegreville, Alberta, Canada. "The Easter Egg Capitol of the World". By Ronald Resch, pioneer in the field of computer art, and member of the Computer Science Faculty at University of Utah from 1970-1979. The programmer that worked with Resch was Robert McDermott (who got his Ph.D. from the work at U. of Utah).

Frank Crow (University of Utah Ph.D. 1975) Developed anti-aliasing methods for edge smoothing.

1974

The first ACM/SIGGRAPH conference is held in Boulder Colorado. There are 600 attendees.

The New York Institute of Technology Computer Graphics Laboratory (CGL) is founded in 1974 Dr. Alexander Schure, and hires recent Utah graduate Edwin Catmull to head the new CGL group. (See the companies chapter for a good history of the NYIT CGL.)

Phong Bui-Tuong develops the Phong shading method at Utah. (Later become a professor at Stanford? When?)

Dr. Ivan Sutherland and associate Glen Flex start a Hollywood company called Picture Design Group with John Whitney Jr. and Gary Demos. One of the first tests they do is for a feature film proposed by Walter Films and Carl Sagan, called *1Cosmos*. Using an E&S Picture System at UC San Diego Demos began tests on one-million-star galaxy simulations. Operating with a clunky front-end system that crashed every fifteen minutes, it forced him to wait 5 minutes to boot, and took 5 minutes to back up data after only 5 minutes of working before the system would crash again. (in addition to having to go so far as to write his own random number generator) They did other work for educational films, and the Museum of Science and Industry, but after about 9 months Ivan wanted to give in favor of going back to academia. Demos and Whitney would then go to Triple-I.

1975

ACM/SIGGRAPH in Bowling Green, Ohio with 300 attendees.

Hunger by Peter Foldes: "First fully animated figurative film every made using computer techniques. (Computer Interpolation or inbetweening). Like Csuri's work, some of the first geometric interpolation or "Morphing" techniques. Foldes would also create the film "Metadata"

The venerable icon of early computer graphics, the famous *1Utah Teapot*, is designed by Martin Newell at the University of Utah.

The TWEEN animation system is developed by Dr. Edwin Catmull at NYIT. Originally written in assembler language (Ed hated Fortran), TWEEN was re-written completely in C to run on UNIX about a year later (It took up 2 megabytes of memory on a PDP-11). He then actually renamed the program *1MO-TRUCK*, for *1motion trucking-thru-the-frames*, but no one would use the new name so TWEEN it stayed.

After 20 years of research Dr. Benoit Mandelbrot publishes his seminal paper: "A Theory of Fractal Sets." The study of fractal geometry is revealed to the popular press. (The theory had been around before, and contributed to by noteworthy mathematicians such as Julia, Poincare, and Falconer. Mandelbrot gave it a name and codified it.)

John Whitney Jr. and Gary Demos form the Motion Picture Project Group at Triple-I.

1976

ACM/SIGGRAPH in Philadelphia, Pennsylvania with 300 attendees, and the first exhibition (with 10 exhibitors!)

Future World: Gary Demos, John Whitey Jr. and a team at Triple-I creates the first feature film appearance of 3D CG; a 3D polygonal representation of a hand, and of actor Peter Fonda's head. (Rendered and filmed out at 2000x2560 pixel resolution.) The film also featured the first ever digital

composite, a sequence of 1samurai warriors, materializing in a chamber room.

Warner Communications buys Atari from Nolan Bushnell for \$28 million

Nelson Max's sphere inversion film shown at SIGGRAPH

Jim Blinn environment (reflection) mapping while a graduate student at the University of Utah. The paper is co-authored with his professor Martin Newell, published in the Communications of the ACM in 1976.

[SIDEBAR] Close Encounters CG\$ALMOST!

Bo Gehring, founder of Bo Gehring Associates of Venice, California, produced computer animation tests for Steven Spielberg's CLOSE ENCOUNTERS OF THE THIRD KIND. Like similar tests created by Triple-I, the tests did not result in any CG production work on the film.

1977

ACM/SIGGRAPH in San Jose, California with 750 attendees and 38 exhibitors.

Star Wars (Twentieth Century Fox)

The Death Star simulation was designed and created by pioneering algorithmic artist Larry Cuba. George Lucas was impressed both by Cuba's early abstract CG film First Fig(1974) and the fact that he had worked with another pioneer of motion control and computer graphics John Whitney Sr. Ben Burt, the films sound designer, had been tasked to get the word out around town and track down bids for the work.

Cuba designed storyboards from the description of the scene in the script, and worked on the job at the University of Illinois Chicago. A 2D drawing program that Cuba designed with the GRASS language was modified to allow input of a third Z axis for every point entered on the digitizing tablet, creating the 3D representation of the Death Star surface.

Using the Vector General based GRASS graphics system designed by Tom DeFanti, Larry worked night and day for 12 weeks to produce 2 minutes of film of which 40 seconds appeared in the final film sequence.

[RENDER TIME QUOTE] 1(While the GRASS system was capable of real time animation) the real time capability came from the Vector General's hardware implementation of basic transformations, like translation, rotation and scaling. also the projection transformation that turns a 3D object into a 2D drawing, but it was only capable of a parallel projection (that is, no 'true perspective'). Since I needed perspective for this project, I was back to using software for the projection and therefore *not* able to animate the scene in real time. I was getting a frame rate of about two minutes of computation per frame and so the whole shot took about 12 hours. Larry Cuba

(A rented Mitchell camera filmed the imagery off of the computer monitor) The finished footage was originally intended to be shot as a rear projected element live on stage with the actors in London, but greatly reduced production deadlines made that impossible. The full story as told by Larry Cuba himself:

[QUOTE] Around two months from my deadline, I was sent a production schedule and I noticed that the live action shooting of the shot that my work was to be used in, was scheduled a month earlier than the delivery date specified on my contract. So instead of having two months left to finish the shot, I had only one.

When I mentioned this to the Assistant Producer, he informed me that it was even worse than that because they required the film to be delivered 4 weeks earlier than shooting in order to have time to make back up copies (should anything happen to the footage during the live action filming). So apparently, since I couldn't send out the shot immediately, we were already dead in the water.

The 'solution' he came back to me with was that they would rearrange the schedule and place that scene (the briefing room scene) on the last day that they had the large sound stage (they were shooting in England. All communication went from me, in Chicago, to the Assistant Producer in LA, to the Producer, Gary Kurtz in London and then to Lucas and then the reverse trip back). This would give me four more weeks to produce the shot (rather than the eight that I thought I had).

So with my schedule cut in half, I stepped up production. I was getting three hours of sleep a night by sleeping on the sofa in the (over air conditioned) lab with the computers. computers generate a lot of heat so computer rooms need to be kept cool or the computers will fail to work.

Working in this way, I was able to finish building the computer model of the Death Star and program the fly through sequence just in time for it to be filmed and sent off. But once I started the film run (which had to run continuously for 12 hours), the computer would crash about 30 minutes into it. Up until this point, the occasional crash was not a problem. reboot and keep going. But now this was a disaster. I couldn't put the shot together filming in 30 minute bursts. (I could if I rewrote the program, but there was no time for that now). We tried everything we could think of to get the system to stop crashing. (we even took the hard disk apart and cleaned it), but 30 minutes after every start, the system crashed.

It was getting late on Saturday night and I had to put the exposed film in the mail on Monday. By 3am (my bedtime), I decided that it was useless. On Monday morning, instead of sending out the film, I would have to call LA and tell them that I had failed to deliver and that our only recourse at this point was to shoot the scene blue screen and optically print my animation in later.

Since there was no more hope, I figured I would at least be more comfortable, so before I went to sleep, I turned off the air conditioning so I wouldn't freeze, and I started the shot from the beginning one more time (what the heck?). This time it ran continuously throughout the night and Sunday morning, completing the shot just in time. Larry Cuba

There was traditional hand animation done for the final four seconds of the bomb entering the death star exhaust port and exploding; completed by John Wash at Image West.

Other computer graphic and video display images were created for Star Wars by several different people. John Wash, Jay Teitzell and Dan O'Bannon at Image West created many electronic video graphic effects for the targeting computers and background tactical displays. Larry Cuba also completed several graphics seen in the DeathStar guard room when R2 and C3PO first tap into the central computer.

[SIGGRAPH FACTOID] The 1977 SIGGRAPH convention Electronic Film Show also ended with Larry Cuba's work,

although not as planned. Halfway through his film *First Fig*, all the power went out in the hotel bringing it, and the show to a premature ending.

1978

ACM/SIGGRAPH in Atlanta, Georgia with about 1500 attendees and 44 exhibitors.

Jim Blinn produces the first in his series of animations for the *The Mechanical Universe* while at JPL.

Jim Blinn also publishes his technique of bump mapping, completed as part of his graduate thesis at the University of Utah the previous year. His demonstration of the new shading code is shown as 128x128 resolution, 16 frame loop of a bumpy sphere. His initial method of calculating both the angle and amount of perturbation is later refined and simplified as an altitude description, allowing for incremental gray scale values to define intermediate angles of surface normals.

1979

ACM/SIGGRAPH in Chicago, Illinois with about 3000 attendees and 79 exhibitors.

Edwin Catmull leaves NYIT to head the Lucasfilm Computer Development Division. He is soon joined by Alvy Ray Smith, David Di Francesco, Tom Duff and Ralph Guggenheim.

[QUOTE] In 1979, the most significant artistic event of my career occurred: Ed Emshwiller and I created *Sunstone*. It is primarily his piece, but we worked very closely on this piece and I am still extremely proud of it. It is in several museum collections of the world, including MOMA. Lance Williams and Garland Stern also helped some on it. Alvy Ray Smith

The Black Hole (Disney): Opening grid/black hole simulation. By John Hughes (Rythm and Hues) et al. at Robert Abel & Associates.

Jim Clark designs his *geometry engine*, the basis for his future company Silicon Graphics.

Alien: Alan Sutcliffe at Systems Simulation Ltd. Of London created a computer monitor sequence showing a 3D terrain fly-over, rendering computer-generated mountains as wireframe images, with hidden line removal.

Meteor has vector graphics created by Triple-I

Julien Gomez develops *TWIXT* at Ohio State software used at Cranston Csurí Productions.

Raytracing developed at Bell Labs & Cornell University. Turner Whitted published a paper for SIGGRAPH 79 describing raytracing techniques.

1980's

The first digital computers used in CG were those in the Digital Equipment Corporation (DEC) line including the early PDP-1, PDP10 and PDP-11 of the last decade. However

because of their cost and high maintenance, these were restricted to large budget University and major production settings. Typical of this work is Jim Blinn at JPL creating the Voyager Flyby films, the Cosmos Series for Carl Sagan, and the Mechanical Universe project; all from about 1979 to 1983.

The workstations, as we know it today were introduced in the early 1980s by companies such as Apple Computer and Silicon Graphics Inc.

The consumer market for personal computer graphics began with the Macintosh personal computer and its MacDraw and MacPaint software in 1984. The Xerox Alto did of course pre-date the Mac by a decade, but did not reach personal use in any numbers; its initial market was government and University settings.

Commercial CG production was boosted by the new generation of digital machines such as the (MORE INFO!) and the early Silicon Graphics workstations such as the IRIS 3130 in 1989. At the same time, third party companies began providing specialized software to run on these new graphic platforms. For 2D graphic design and image processing, Photoshop was introduced for the Mac in 198?. Early 3D animation software for the higher end market included Wavefront(1987), Intelligent Light(198?), and Alias v1.0(1984).

The mid 1980s to early 1990s were a time of tremendous advances in technology and stunning creative breakthroughs.

Companies such as Robert Abel and Associates, Triple-I, Magi/Synthavision, Omnibus, and Digital Productions created such memorable images as Sexy Robot (ABEL), Chromosaurs (PDI), and the Benson & Hedges(Digital Productions) commercials.

The U.S. National Science Foundation began to provide supercomputer access to university research programs, including the University of Illinois Supercomputing Center.

1980

ACM/SIGGRAPH in Seattle, Washington with about 7500 attendees and 80 exhibitors.

LOOKER: Triple-I produces seven minutes of computer graphics under the Direction of Richard Taylor et al. Polygonal models of a complete human body were created.

Loren Carpenter's fractal extravaganza "Vol Libre" is presented at SIGGRAPH 80

Loren Carpenter at Lucasfilm's Games Group & Atari created "Rescue From Fractalus!"

Chris Briscoe and Paul Brown co-founded Digital Pictures as the UK's first specialist computer animation company

1981

ACM/SIGGRAPH in Dallas, Texas with 14,000 attendees and 124 exhibitors.

Nelson Max begins making computer graphics for the IMAX film format at Lawrence Livermore National Labs. Steve Levine and George Matthews here also had lots of contact with NYIT in the early days. They were making graphics of "superheated spheres" (get it?)

Computer Assisted Animation Stand(CAAS) at NYIT Computer Graphics Lab.

Omnibus Video Inc. is founded in Toronto Canada.

Adam Powers (The juggling tuxedo guy): Part of Information International Inc. (III) demo reel shown at SIGGRAPH that year.

Nintendo introduces the Donkey Kong video game

1982

ACM/SIGGRAPH in Boston, Massachusetts with about 17,000 attendees and 172 exhibitors.

Tom Brigham (NYIT) introduces the first full raster morph technique at the 1982 SIGGRAPH conference.

Silicon Graphics Inc. formed by Jim Clark (University of Utah 197?) For lots of details see the Companies chapter.

Autodesk formed by Dan Drake and John Walker, release AutoCAD v1.0 at COMDEX.

Mits Kaneko and the Japan Computer Graphics Lab (JCGL) produce the series "The Yearling". Episode No. 2 was broadcast in April 1982 and became the world's first television animated program completely processed with a computer. (See the Company history on JCGL for more details.)

The first all digital computer generated image sequence for a motion picture film: Star Trek II: Wrath of Khan/genesis sequence. Amazing use of fractal geometry and particle systems, (by Loren Carpenter based on his own work from his Vol Libre film, completed while at Boeing). Bill Reeves fire, Tom Porters stars, and Tom Duffs moon. Conceived and Directed by Alvy Ray Smith.

Tron (Disney)

The first extensive use of 3D CGI animation for a feature film. This milestone project was originally boarded by Bill Kroyer and Jerry Rees and pitched to Disney by Steve Listberger. Bill and Jerry came up with the titles Computer Image Choreographers for their roles which were much more than traditional Animation Directors. The model motion and choreography, along with camera blocking and motion paths were all sketched out in exacting detail to be passed on and realized precisely by four CG production houses.

[TRON FACTOID] The largest format pencil tests ever! The Disney art and animation team that were previsualizing the CG for the film, never had any way to view a traditional pencil test. The first time they got a chance to see their planned motion scenes was only after the CG was created, rendered and output to 70mm film. Because of a technical limitations at Disney, the film was actually rear projected in the screening room.

So who did what CG on TRON? Robert Abel & Associates created the title sequence for the film, and the entry to the digital computer world. Digital Effects created the little bit character. Mathematical Applications Group Inc. (MAGI) created the light cycles and most of the recognizers. Information International Inc. (Triple-I) created Sarkas carrier, the solar sailer, and the MCP character sequences near the end of the film.

In total, there was actually only about 15 minutes of computer generated imagery created for the film, supervised by Richard Taylor. The majority of effects were accomplished by traditional animation techniques involving tens of thousands of hand rotoscoped individual frames of artwork.

1982/83

Where the Wild Things Are (Test done at MAGI): The first instance of digital compositing for motion picture work. The character animation was done at Disney (lead by Glen Keane,) and the cg backgrounds, rendering, painting, and compositing was done at Magi/Synthavision. Jon Lasseter was the official Disney-Magi liaison. Ken Perlin supervised the project, with the CG work lead by Chris Wedge and Jan Carlee (both now at Blue Sky.). Software was by Ken Perlin, Christine Chang, Gene Miller, and Josh Pines. Look for many more details in the Companies Chapter!

1983

ACM/SIGGRAPH in Detroit, Michigan with about 14,000 attendees and 195 exhibitors.

AVCO Finance spot shown at SIGGRAPH Electronic Theatre. (This was the first fully rendered raster 30-second commercial spot.)

Alias Research Inc. founded in Toronto Canada

The Bosch FGS-4000 (the first true turnkey 3-D System) is introduced at NAB in 1983.

Cube Quest(Simutrek Inc.): Early 3D graphics video game.

Return Of The Jedi (Twentieth Century-Fox/LucasFilm Ltd.): Holographic Endor moon sequence by the LucasFilm Computer Graphics Group. Bill Reeves and John Lasseter did it using vector graphics to simulate raster graphics!

1984

ACM/SIGGRAPH in Minneapolis, Minnesota with 20,390 attendees and 218 exhibitors.

Synthavision, a division of MAGI, is sold off to a Canadian investment company.

Silicon Graphics releases its first commercial product, the IRIS 1000 terminal (which ran off a VAX host).

Wavefront software company formed in Santa Barbara, CA by Bill Kovacks et al & LOTS MORE

A modern global illumination rendering technique called Radiosity is presented by a team led by Don Greenberg at Cornell University.

The Apple Macintosh computer is released. The first personal computer with a graphical user interface (GUI).

The Adventures Of Andre And Wally B. LucasFilm Computer Graphics Division. Alvy Ray Smith directed John Lasseter in his first CG short animated film.

[SIDEBAR NOT!] Dune: Cool 3D CGI body armor. NOT! (Traditional animation done by Jeff Burks while at Van derVeer Photo Effects.)

The Last Starfighter (Lorimar): The first CG project by the new Digital Productions formed by Gary Demos and John

Whitney Jr. after having just left Triple-I.

2010: Odyssey Two: Digital Productions worked with Boss Film Corp. as Richard Edlund, Larry Yaeger, Craig Upson, Neil Krepela, et al. combined computational fluid dynamics with CGI to create the planet Jupiter.

1985

ACM/SIGGRAPH in San Francisco, California with 27,000 attendees and 254 exhibitors.

Disney's The Black Cauldron is the first use of 3D computer graphic elements in an animated film. (true?)

The first ever Academy of Motion Picture Arts and Sciences award recognition for computer graphics achievement: John Whitney Jr. and Gary Demos of Digital Productions receive The Scientific and Engineering Award was for the practical simulation of motion picture photography by means of computer generated images (1984).

Bob Abel's Sexy Robot completed for the Canned Food Council.

The animated short film Tony de Peltrie by Phillippe Bergeron shows at SIGGRAPH 85. Using digitized clay models, and the new user friendly TAARNA 3D animation system (From U. of Montreal) along with additional keyframe interpolating algorithms by Doris Kochanek described at the previous years SIGGRAPH. (Phillippe also did hero animation on the Symbolics short Stanley and Stella in 1985)

[SIDEBAR NOT!] Max headroom was NOT computer generated. (Really, take my word for it.) Beginning with the 1985 British music video show and TV pilot, he was portrayed by actor Matt Frewer in stylized makeup with added video editing effects. The US TV series produced in 1987 did feature some other on screen CG (created with an Amiga) but never Max himself. (BTW, 10 years later actor Matt Frewer later starred in the LawnmowerMan II sequel infinitely less good than Max IMO) For all things Max visit:
<http://www.maxheadroom.com/altfaq.html>

Commodore introduces the Amiga color personal computer.

Playland (Atari Corp.): Bill Kovacs.

Los Alamos National Lab: The Ultra-High Speed Graphics Project is started. It pioneers animation as a visualization tool and requires gigabit-per-second communication capacity.

An early massively parallel (128-node) Intel computer is installed.

Young Sherlock Holmes: The stained glass knight sequence.

€The first CG Character in a feature film

€The first computer generated images in a feature film to be exposed directly onto the film with a laser.

€One shot was also the first ever all digital composite of CG with live action footage for a feature film. (The rack focus shot that starts on the knight's hands grasping the sword hilt and then tilts up to his face)

By the graphics group at LucasFilm LTD.

[FACTOID] David DiFrancesco built the digital film printer, that was used for Young Sherlock Holmes. Designed as one unit with three main components; a scanner and a printer with a Pixar Image Computer in between. The former video artist would later receive two separate Academy Awards for

his pioneering work. A Sci-Tech Award in 1994 for the scanner portion, and a Technical Achievement Award in 1999 for the printer work.

Money For Nothing MTV video by Dire Straits. (Steve Barron director) Gavin Blair and Ian Pearson created the animation at Rushes Post production in London, done on the Bosch FGS-4000. The Quantel effects were done by Viv Scott. Ian and Gavin now own and run a company in Vancouver called Mainframe, out of which they produced Reboot(1994).

Cranston-Csuri produces many national broadcast network graphics, but closes in 1987. Many of its employees go on to later form MetroLight Studios (1987).

[BIO] Gary Demos: (studied under Ivan Sutherland at Utah?) Š Cal Tech, went to work at E&S in 1972 and met John Whitney Jr. Began working on projects with III then went with Whitney to III to form the IMotion Picture Design Group, in 1974. Left III just before Tron production, again with Whitney, to form there own company Digital Productions. DP filed for chapter 11 in 198? But was then continued as Optimistic by Whitney. Demos the formed his own company, which still exists today: DemoGraFX.

1986

ACM/SIGGRAPH in Dallas, Texas with about 22,000 attendees and 253 exhibitors.

SoftImage founded in Montreal by Daniel Langois.

Mick Jagger's Hard Woman music video. Digital Productions Brad deGraf, Bill Kroyer, Kevin Rafferty. Et al. CG Co-Produced by Nancy St.John and Alan Peach.

"The Juggler": An Amiga demo by Eric Graham.

Digital Productions create the three minute opening sequence for the feature film Labyrinth.

Complex 2D vector graphics character animation was produced by Digital Productions for the Mick Jagger music video Hard Woman.

PIXAR formed by Lucasfilm Computer Graphics Division pioneers Edwin Catmull and Alvy Ray Smith along with about 35 others including John Lasseter, Ralph Guggenheim, Bill Reeves, et al. Purchased from George Lucas by Steve Jobs (Apple/NeXT) for \$10 million.

Luxo Jr. (PIXAR Animation Studios): First CG Short Animated Film to be nominated for an Oscar for Best Short Animated Film

Flight of the Navigator: Omnibus Computer Graphics creates the silvery reflective spaceship. Contributors included Jeff Kleiser(KWCC), Les Major(ILM) and Kevin Tureski(Alias)

The Great Mouse Detective: Disney first use of 3D computer graphic elements in an animated film. (Or was it The Black Cauldron in 1985?)

Howard the Duck: first digital wire removal for a feature film. Painted by Bruce Wallace at ILM with proprietary

1 Layerpaint, software on a Pixar Image computer. Layerpaint code originally written by Mark Leather and modified by Jonathan Luskin and Doug Smythe.

Star Trek IV: First use of Cyberware 3D scanner for film

Digital Productions is purchased (June), then also Robert Abel & Associates (September), by Omnibus Computer Graphics in 1986. Omnibus goes out of business one year later on April 13th 1987.

1987

ACM/SIGGRAPH in Anaheim, California with about 30,541 attendees and 274 exhibitors.

Rhythm and Hues formed by ex-Abel staffers, opens in a former dentist office.

Captain Power and the Soldiers of the Future

The first television series to include 3D characters that were done entirely with computer animation. It went on the air (September) in North America. Soaron and Blastarr were two CG robots that appeared in the 22 episode series. The computer animation was produced by Arcca Animation in Toronto.

[SYNTHAVISION FACTOID] 1 Arcca was the reformation of Sythavision staff and software to do the Captain Power series that was a creation of Landmark Entertainment (Hollywood) and financed by Mattel. The show featured toys that were interactive with the television show by registering blast hits on the toy (via a 30hz flicker on TV) or on the TV show character (via a trigger pull during a 15hz flicker from the TV). Paul Griffin

About four minutes of computer graphics was animated for each episode every week using two SGI 3130 workstations running Wavefront software. The motion was then ported over to Sythavision data.

[ANIMATING WITH STICKS AND STONES] 1 Animation was incredibly arduous sometimes. First you'd plot the model and the path of your animation on graph paper. Then input hundreds or thousands of text lines in a form that Sythavision would understand. If you were out as much as a space or tab in your input file, it wouldn't run. To review your animation, you played in back by flipping images through a frame buffer that often time had pixels as big as postage stamps and based on this make a decision as to whether or not to send your rendered animation to the film recorder. Two days later it would come back from the lab and you could see where all the mistakes were and start over again. But it was a beautiful renderer. The quality of the solid modelled surfaces and the lighting routines made for some great images. Paul Griffin

Rendering was done on 13 Sun Workstations that ran a proprietary job control system, that would pick up new frames in a sequence as they were completed, which may have been the first render farm of its time. The work for the show won

Arcca a Gemini Award (the pinnacle in Canadian film production) for Technical Achievement in 1988.

The producer was Bob Robbins. The art director was Earl Huddleston. Paul Griffin (ILM) was Animation Director, Andy Varty, Sylvia Wong (Rhythm & Hues, ILM), Les Major (ILM, Pixar). Paintbox work by Rob Smith and Mike Huffman. Jenniffer Julich was in charge of storyboards. Rob Coleman,

was Arcca's onset liason/line producer. Mark Mayerson now directs Monster by Mistake on DisneyTV and YTV (Canada). On the live action production side, Doug Netter (Rattlesnake Productions) and Larry Dittillo(sp?) (the writer) went on to develop Babyon 5.

1988

ACM/SIGGRAPH in Atlanta, Georgia with about 19,000 attendees and 249 exhibitors.

Fruit Machine (Wonder World): The first all digital film composite for a feature film outside the U.S. by Computer Film Company (CFC)/London. Multiple film elements were scanned into a computer, 100% digitally composited, and filmed back out again.

see the Companies chapter on CFC for more details

Jim Henson and Digital Productions create a real-time 3D digital character for the Jim Henson Hour. The first of its kind. Steve Whitmeyer(sp?) was the puppeteer and voice. Thad Bier(PDI/Hammerhead) and Grahm Walters and Rex, shipped all the equipment up to Toronto one week before SIGGRAPH. The opening to the show was done by Jamie Dixon(PDI/Hammerhead).

Mike the Talking Head The first real-time character (aka motion-capture, vactor, performance animation). Michael Wahrman and Brad deGraf did it at deGraf/Wahrman live at the SIGGRAPH Electronic Theatre in Atlanta. (Mike was a virtual caricature of the late Mike Gribble, the host of that show, and the Mike of Spike and Mike's animation festival.)

Willow (MGM/Lucasfilm Ltd.): First feature film use of digital morphing technology.

CAPS(Computer Animation Paint System) developed jointly between Pixar and Disney.

Tin Toy (PIXAR Animation Studios): First CG Short Animated Film to win an Oscar for Best Short Animated Film

1989

ACM/SIGGRAPH in Boston, Massachusetts with 27,000 attendees and 238 exhibitors.

Indiana Jones and the Last Crusade (Lucasfilm Ltd. /Paramount): Contrary to what you may have read elsewhere, this was NOT he first all digital composite for a feature film. (ILM's own Stained Glass Knight in Young Sherlock Holmes, CFC's Fruit Machine, and Triple-I's Future World all came before) The 1Donovan's destruction sequence by ILM was the first to use many multiple scanned film elements, digitally composited, and then scanned back out to film with a laser.

(By now it gets a little silly with all of the sub-sub classifications of 1firsts, in areas such as these.)

The Abyss (GJP Productions/Twentieth Century-Fox): Water Pseudopod.

1990's

The entertainment world as we know it began to change in the 1980s when motion picture images in Tron, Star Trek II, The Last Starfighter, and Young Sherlock Holmes gave the

audience a taste of the future. Now, George Lucas's Industrial Light + Magic began to continuously raise the popular standard by which all CG was judged by creating such images as the water pseudopod in James Cameron's film The Abyss (1989) and the T-1000 in Terminator 2: Judgment Day (1990). In 1993 ILM smashed all previous conceptions about computer graphics when Jurassic Park's photo-real dinosaurs took center stage in theaters around the world.

1990

ACM/SIGGRAPH in Dallas, Texas with 24,684 attendees and 248 exhibitors.

The feature film *Flight of the Intruder*. Rhythm and Hues created over 30 shots of photo-realistic aircraft, cluster bombs, and smoke in full daylight..all with their own proprietary software.

deGraf/Wahrman did The Fantastic World of Hanna-Barbera, the first CG ridefilm. It was a fully 3D chase/ride through Bedrock and Scooby-Doo's castle, with cel animated characters, for Universal Studios Florida. (Additional CG work by Rhythm and Hues)

Robocop 2 (Also by deGraf/Wahrman) was the first use in feature films of Performance Animation. Among those who also contributed were Ken Cope(animation) and Gregory Ercolano(TD).

Kroyer Films creates the full length animated feature film FernGully: The Last Rainforest. It contains 40,000 3D hidden line computer plotted cel frames to augment the bulk of the traditional animation. It also contains a digital-ink-and-paint sequence by Sydney-Right, a feature film first.

The Rescuers Down Under: The first complete feature film to be *completely* digital. The CAPS system digitally ink and paints every frame of the film.

Die Hard 2: Die Harder (Twentieth Century-Fox): The first digitally manipulated matte painting created at Industrial Light & Magic. Matte department supervisor was Bruce Walters, Paul Huston and Michael McAllister helped in design and composition and Yusei Uesugi was the matte painter extraordinaire. Four separate images were digitized from the painting (13 feet wide by 5 feet tall), decreasing in resolution from the center outward. The images were assembled in a MacII computer, and manipulated by Uesugi using Photoshop. The image was combined with numerous live-action elements of people, lights and steam with a camera move programmed by Pat Myers.

NewTek releases the Amiga based Video Toaster.

1991

ACM/SIGGRAPH in Las Vegas, Nevada with about 23,100 attendees and 282 exhibitors.

Terminator 2: Judgement Day (Carolco): T-1000 liquid metal cyborg

Beautiful all CG commercials by PIXAR for Listerine, Life Savers and Tropicana set a new standard for broadcast excellence.

Disney's Beauty And The Beast ballroom sequence is a major new direction in feature length animated films.

1992

ACM/SIGGRAPH in Chicago, Illinois with 34,148 attendees and 253 exhibitors.

Death Becomes Her (Universal): Photoreal human skin and body replacement.

1993

ACM/SIGGRAPH in Anaheim, California with 27,000 attendees and 285 exhibitors.

Wavefront acquires the TDI software company from Thompson Corp of France. In exchange Wavefront receives a major capital investment from Thompson

PDI opens a Hollywood production office. This office would close in a short few years.

Marc Scaparro, Eric Gregory and Brad deGraf did Moxy for the Cartoon Network at Colossal Pictures. Produced by Anne Brilz. It was the first live broadcast of a virtual character.

Jurassic Park (Amblin/Universal): Photo-real 3D Digital Dinosaurs

1994

ACM/SIGGRAPH in Orlando, Florida with about 25,000 attendees and 269 exhibitors.

Reboot: the first 100% CGI television series airs on ABC from Mainframe Entertainment Inc.

Microsoft acquires Softimage

Forrest Gump (Paramount): Photoreal/invisible 3D and 2D digital effects blending new footage with old, changing archive footage, and removing Gary Sinese(sp?) legs! By ILM of course.

Flintstones (Universal): First feature film digital hair developed for the saber toothed tiger.

1995

ACM/SIGGRAPH in Los Angeles, California with 40,100 attendees and 297 exhibitors.

Silicon Graphics, Inc. acquires both Alias and Wavefront, merging the two companies.

Toy Story (PIXAR Animation Studios): First full length CG Animated feature film. Director John Lasseter wins a Special Achievement Academy Award.

Judge Dredd (Cinergi): Early examples of fully 3D digital stunt people by the Kleiser-Walzack Construction Company for Mass-Illusion.

Casino (Dir. Martin Scorsese): Matte World Digital utilizes LightScape software to seamlessly integrate a 1970s virtual Las Vegas strip into present day live action footage. The first time radiosity lighting was used in a feature film.

Batman Forever (Warner Brothers): Early example of 3D realistic digital stuntman by Warner Brother Imaging Technology (W.B.I.T.) and Pacific Data Images. Also a very realistic, fully 3D cityscape by W.B.I.T.

Casper (Amblin/Universal): Record number of on screen shots with a digital character. 400+

Jumanji (Tri-Star): Further development of particle based digital hair technology for Lion sequence.

1996

ACM/SIGGRAPH in New Orleans, Louisiana with 28,800 attendees and 321 exhibitors.

Alvy Ray Smith, Ed Catmull, Tom Porter, and Tom Duff receive a Technical Academy Award for digital image compositing (ie the alpha channel)

Dragonheart (Universal): Breakthrough 3D CGI character animation and lip-synch dialog.

Twister: Breakthrough realistic tornadoes and weather effects by Industrial Light and Magic using Wavefront's Dynamation..

1997

ACM/SIGGRAPH in Los Angeles, California with 48,700 attendees and 359 exhibitors.

Floops (done at Protozoa by Brad deGraf, Emre Yilmaz, Steve Rein and others) was the first character distributed as 3D (VRML), the first episodic cartoon on the Web, and the first significant animation on the web (30 minutes worth).

Star Wars/Special Edition (Twentieth Century-Fox/LucasFilm Ltd.): Restored and enhanced 20 year old film footage. About 350 shots were added or modified for all three films.

Spawn: Photo-real fully 3D creature transformations, full screen digital stunt doubles, and dynamic simulated cape. All with bone-cracking, digital-drool slinging realism.

Titanic: Large scale use of motion-capture and 3D digital crowd extras.

[Quantel loses PATENTS ISSUE]

British company Quantel has been asserting a set of patents against companies for about a decade now, patents that many of us in the digital imaging and computer graphics business believed were invalid. These 1980s-vintage patents covered airbrushing (digital painting with soft-edged brushes), digital image compositing (!), pressure-sensitive stylus, mixing paints on a window called a palette, etc. I and my colleagues have long believed these notions to be too simple to be worthy of "invention" hence patent coverage.

Furthermore, if anyone were to deserve credit for the "invention", it certainly wasn't the Quantel people in the 1980s but rather several of the many practitioners (in the US mainly) in the 1970s.

Jim Blinn, Lance Williams, and I tried to help save a British company called Spaceward from these patents in a London trial in 1989. We were unsuccessful, largely, I believe, because we didn't have any hard evidence - no code, no program. This changed in September 1997. Quantel sued Adobe, well-known US producer of popular software products such as Photoshop, for patent infringement on US versions of the UK patents that had been held against Spaceward. The trial was held in Wilmington, Delaware. The following colleagues joined me in helping Adobe this time: Marc Levoy, Christie Barton, David Em, Dick Phillips, Jim Blinn (by deposition), and others.

The Adobe attorneys did a great job of gathering evidence, including hard evidence this time. They obtained actual code that I had written in 1977 and 1978 and recompiled it under Windows. So I was able to demonstrate directly to the jury exactly what we had done in the 1970s - in this case I was showing the first full-color (RGB, or 24-bit) digital paint program, which of course did many of the things Quantel claimed to have invented in the 1980s! Marc Levoy's 1978 full-color paint program (the second one) was similarly found and recompiled and shown to the jury.

Long story short: The jury found all five patents at issue invalid (and that Adobe was innocent of infringement).

-Alvy Ray Smith

1998

ACM/SIGGRAPH in Orlando, Florida with ??? attendees and ?? exhibitors.

1998 saw an unprecedented number of SciTech awards go to the computer graphics community. Individuals at Alias, Pixar, PDI, Side Effects, SoftImage and Wavefront all were recognized for various components of those systems. In addition, several individuals were recognized for their contributions to CG.

A Scientific and Engineering Academy Award was awarded to Richard Shoup, Alvy Ray Smith and Thomas Porter for their pioneering efforts in the development of digital paint systems used in motion picture production. The award reads:
 Much of the foundation for the numerous contemporary digital paint products for motion pictures can be traced directly back to the early work of these digital pioneers.

A Scientific and Engineering Academy Award was awarded to Craig Reynolds for his pioneering contributions to the development of three-dimensional computer animation for motion picture production. The award reads:
 The early contributions of Mr. Reynolds in the digital animation arena have become both influential and instrumental in the architecture of many later systems developed at companies throughout the computer animation industry.

Geri's Game (Pixar): Academy Award winning animated short film showcases the newly rediscovered modeling technique of subdivision surfaces.

Bingo (Alias|Wavefront) Chris Landreth's test piece for the initial Maya release received a Genie Award from The Academy of Canadian Cinema and Television. It was named Best Computer Animation at Ottawa 98 and at Imagina in Monaco, Bingo also received an award from France's Societe des Auteurs et Compositeurs Dramatiques (SACD) for "Most Innovative Story and Production".

Antz is released (PDI/Dreamworks)

A Bug's Life (Pixar/Disney)

1999

Autodesk merges its newly acquired Discreet Logic (Montreal) division with its Kinetix (San Francisco) division into the new Discreet entertainment division.

May 19th, 1999 Star Wars/Episode 1: The Phantom Menace
Almost 2000 state-of-the-art digital effect shots, most of which are created at Industrial Light & Magic in less than two years production time. The Gungan JarJar Binks is the first all digital leading character in a motion picture.

The few shots that were not effects related were also scanned and color corrected to produce a full digital master.

Later that summer, LucasFilm premieres the film in New York and LA with a new electronic projection system. The Texas Instruments system uses 1920x1080 progressive video resolution to project the film at 24fps directly from digital storage.
